

LESSON GUIDE

Technology Plus

Unit 6: Database

5.1.1 – Data

Lesson Overview:

In this lesson, students will identify data and information as an asset, explore the value data has in our society, understand the difference between critical data and non-critical data, and explain how to manage and protect them.

Students will:

- Identify the value of data and information as an asset.
- Understand the difference between critical data and non-critical data.
- Explain how to manage and protect critical and non-critical data.

Guiding Question: Why is data and information an asset, and what is the difference between critical data and non-critical data?

Suggested Grade Levels: 8-12

Technology Needed: No

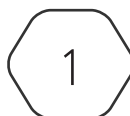
Approximate Time Required: 45-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 5.1 – Explain the value of data and information.
 - Data and information as an asset
 - Critical vs. non-critical data
 - Data-driven business decisions
 - Data capture and collection
 - Data correlation
 - Meaningful reporting
 - Big Data
 - Data monetization
 - Data analytics

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Lesson Background

Background Information:

Data and information are important assets in our society. They shape our daily lives and help us make important decisions and solve problems. Companies and individuals rely on data to improve products, guide strategies, and support success. However, when data falls into the wrong hands it can also be misused by malicious actors to cause harm or steal money. That's why it's important to understand the two main types of data: critical and non-critical, and how to manage and protect them properly.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 5.1.1 – Data • Activity Slides 5.1.1 – Guess Who
Materials per Student	• Student Notes 5.1.1– Data • Student Handout 5.1.1 – Data • Assessment 5.1.1 – Data (optional)

Cyber Terms:

Term	Definition
data	distinct pieces of information, usually formatted in a special way
information	data that has been processed, organized, or structured in a way that makes it meaningful
critical data	essential information an organization needs to operate, meet legal requirements, or protect its customers
non-critical data	information that supports an organization's work but isn't essential for day-to-day operations or legal requirements
personally identifiable information (PII)	data that could reveal a person's identity, includes direct PII and indirect PII
direct PII	information that is unique to one person, like a passport number or IP address
indirect PII	information that is not unique to one person, like a ZIP code or height

5.1.1 – Data

Guiding Question: Why is data and information an asset, and what is the difference between critical data and non-critical data?

Lesson Launch: (5-10 minutes)

Data is the New Oil

- Ask students if they have ever heard the phrase, “Data is the new oil.” Have students think about the importance of oil and how we use it often; guide them to data being similar.
- Share the definitions of **data** and **information**, how they help us, and everyday examples of using data and information.
- Ask students if they have any other daily examples of using data and information not listed on slide 5.

Data vs. Data (15-20 minutes)

- Distribute Student Handout 5.1.1 – Data and have students fill in the answers during the lesson.
- Data as an Asset, give detailed examples as needed.
- Information as an Asset, give detailed examples as needed.

Critical Data vs. Non-Critical Data

- Introduce the two classifications of data, **critical data** and **non-critical data**, and their definitions; emphasize that they are both important to an organization or individual.
- Discuss **personally identifiable information (PII)**, distinguish between direct PII and indirect PII, and how they correlate to critical and non-critical data.
- Discuss business examples of each data class; these are examples that would mostly occur with organizations, elaborate as needed.
- Discuss personal examples of each data class; these are examples that would mostly occur with individuals, elaborate as needed.
- Discuss what could happen to each data class if the wrong person gained access; emphasize that critical data exposure can be harmful, but non-critical data exposure is typically harmless; give examples as needed.
- Discuss management and protection strategies of each data class, point out the differences and similarities. Lead students to discuss why there are some differences between the strategies.

Activity: 5.1.1 Guess Who? (15-20 minutes)

- Display Activity Slides 5.1.1 – Guess Who, have students write their answers for each scenario at the bottom of their handout.
- Decide how you will cover the scenarios; you can read them aloud, call on students to read them aloud, or have students read them independently.
- Provide students with time to write their answers.
- Some scenarios have multiple questions.
- Students may have different ideas about each scenario.

Scenarios and Answers:

- Someone leaked Tech Company's meeting notes online. Are meeting notes critical data or non-critical data? What can the Tech Company do to prevent this from happening again?

Meeting notes are non-critical data. Although this cannot harm Tech Company, they should check their access controls and make sure employees are following basic security principles.

- A stranger called Carlos and told him there was suspicious activity happening with his Social Media Account. The caller asked Carlos for his username and password so they could stop the suspicious activity. Should Carlos share his username and password? Are username and password critical data or non-critical data? What can Carlos do to protect his information?

No, Carlos should not share any information. A username and password are critical data, if this gets into the wrong hands someone could use it to cause harm. Carlos can update his password so it cannot be figured out easily, Carlos could block this number if he is worried about the person calling back.

- A malicious actor gained unauthorized access to Retail Store's database system through an employee. Once inside, the attacker used customers' payment details to buy things online and sell them to others. Are the payment details of a customer critical data or non-critical data? What can Retail Store do to prevent this from happening again?

The payment details of a customer are critical data, someone can use this to take a customer's hard-earned money. Retail Store should implement robust security protocols, like encryption and auditing. They should also make sure employees are compliant with the rules regarding cybersecurity.

- Mary's brother thought it would be funny to delete a playlist she made for her birthday party. Mary could not remember all the songs on her playlist and was very upset. Is Mary's playlist critical data or non-critical data? What can Mary do to prevent her brother from doing this again?

Mary's playlist is non-critical data. Although this cannot cause Mary direct harm, it could tarnish her relationship with her brother. Mary should change the passcode to her phone and complete regular backups on her phone.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the following questions as a class.
- What makes data and information an asset? Why are they important?

Data and information are valuable assets that help with problem solving and decision making in everyday life.

- What is the difference between critical data and non-critical data ?

Critical data is essential for an organization's main operations, legal requirements, or strategic plans. Non-critical data supports an organization's work but isn't essential for day-to-day operations or legal requirements.

- What are some examples of critical data?

Accept any accurate answers that meet the requirements for critical data.

- What are some examples of non-critical data?

Accept any accurate answers that meet the requirements for non-critical data.

- How can we protect both critical and non-critical data?

Critical data can be protected by following robust security measures, backup and recovery procedures, and compliance rules. Non-critical data can be protected by completing basic security practices, regular backups, and appropriate access controls.